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Full Length Article

A Gaussian Process-Based extended Goldak heat source model for finite element simulation of laser powder bed fusion additive manufacturing process☆

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ABSTRACT

Laser powder bed fusion (L-PBF) additive manufacturing (AM) is a key enabling technology to manufacture highly complex and integrated metallic structures. In L-PBF AM process, the melting of the metal powders and the layers underneath can be governed by either “conduction mode” or “keyhole mode”, with the keyhole mode reportedly leading to porosity and decreased strength and ductility by many studies. In part scale simulations, finite element (FE) model is often used to study the temperature distribution during printing and to predict the residual stress, where a volumetric heat flux with a Gaussian or a double ellipsoidal (Goldak) distribution is often applied as the laser heat source. However, the above heat source models can only capture the melt pool shape in the conduction mode, and fail to capture the transition to keyhole melting mode when the process parameters change. To overcome this inaccuracy, an extended Goldak heat source model is proposed by introducing a laser penetration term as a function of laser parameters obtained from a Gaussian-Process (GP) model. The model is validated by “2D pad” AlSi10Mg L-PBF experiments under a wide range of laser power, scan speed, and laser focus offset, and the results show the model successfully captures the measured melt pool shape in all conditions.

1. Introduction

Laser powder bed fusion (L-PBF) additive manufacturing (AM) is a key enabling technology to manufacture highly complex and integrated metallic structures in automotive industry [1–3]. In L-PBF AM process, the melting of the metal powders and the substrate can be governed by either “conduction mode” or “keyhole mode”, as illustrated in Fig. 1. In the conduction mode, the melt pool is stable and temperature distribution is controlled predominantly by thermal conduction, resulting in a semicircular melt pool shape[4]. As the laser energy density increases, the temperature within the melt pool exceeds the boiling temperature, and metal evaporation occurs under the laser spot leading to a recoil pressure, which depresses and push backward the underneath melt flow and forms a deep keyhole shape melt pool [4–8]. Many studies have reported that the keyhole mode in the L-PBF process leads to manufacturing defects, because the intense and unstable melt flow forms vortex at the rear part of the melt pool which traps bubbles and leads to porosity in the solidified microstructure, leading to decreased mechanical performance [9]. It is therefore important to avoid the transition from conduction mode to keyhole mode in L-PBF process by choosing the appropriate control parameters.

In L-PBF simulations, the key influencing parameters, such as laser power, scan speed, and laser defocusing distance (i.e., spot size) are applied to the numerical model as laser heat inputs. Choosing the appropriate laser heat source model is therefore critical for capturing the keyhole mode melt pool. Depends on the modeling lengthscale and targets of investigation, the simulation models for L-PBF process can be generally divided into continuum-scale and particle-scale methods. The

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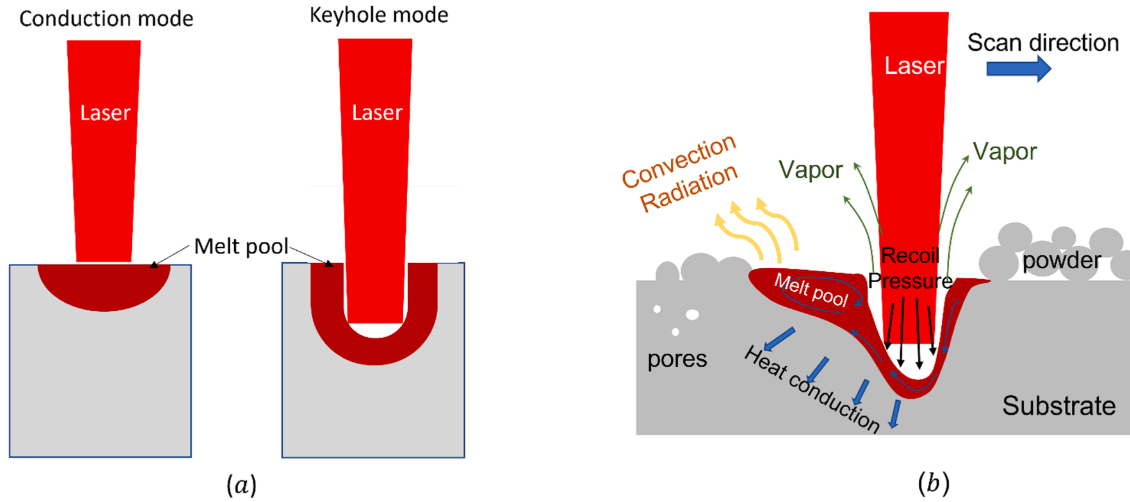

Fig. 1. (a) Brief schematic comparison of conduction melting mode and keyhole melting mode (b) Schematic diagram explaining keyhole formation phenomena during L-PBF process.

continuum scale models simulate multiple tracks and/or multiple deposition layers and often are used for studying part-scale residual stress and thermal history which are influenced by laser scan path strategy, part geometry and boundary conditions. In contrast, the particle scale models usually simulate single track or limited number of tracks and focus on investigating the physics of laser-particles interaction, melt pool peak temperature and dynamics, recoil pressure, evaporation and etc. Because of the fundamental difference at the two lengthscales, the laser heat source models for the continuum-scale and particle-scale simulations are different too. In continuum-scale simulation, the laser heat is applied using a surface or volumetric power-distribution function[10,11]; while in particles-based simulations, the ray-tracing models [12–14] or beam trajectories-based physics-informed models [15,16] are gaining popularity in addition to conventional volumetric distribution models.

The target of this paper focus on improving the laser heat source model for continuum-scale simulation, especially for part-scale simulations. For the apparent reasons, the explicit ray-tracing and laser-particles interaction are not considered in this scenario. The continuum-scale L-PBF process simulations are based on FE methods, where the laser energy input as a surface or volumetric heat flux based on a Gaussian or double ellipsoidal shape (Goldak) distribution. In some simpler models, the laser heat are introduced as an uniformly distributed heat source over a cylindrical or conical shape region [17,18]. It is noted that in continuum-scale L-PBF simulations, the effect of laser-powders interaction is simplified and implicitly considered using one of two methods: either considering the effects of powders in the physical properties, or considering the powder effect in the heat transfer. In the first method, the properties of an element (e.g., density, thermal conductivity) are functions of temperature to account for the phase change [19]. In the second method, the laser beam power density is modified by considering the sheltering effects of flighting particles [20].

Recent studies by Zhang et al. [21] on the existing volumetric heat source models concluded that they are only suitable for conduction mode melt pool formation, and that they fail to capture the keyhole mode high depth-to-width ratio melt pool that was observed experimentally. The lack of predictive capability for the keyhole mode limits the applicability of FE thermal simulations, particularly when the part geometry or scan path is complex. Several methods have been developed to improve the heat source model. Zhang et al. proposed a method to model keyhole melt pool in L-PBF process by modifying the local thermal conductivity matrix and the laser energy absorptivity [21]. Lorin et al. [22] developed a laser heat source model for FE simulation of keyhole mode laser welding by combining a surface Gaussian heat source and a conical heat source inside the material. Flint et al. [23] proposed an extended Goldak heat source model for laser welding by introducing a double-conical heat source function. Several new model parameters were also introduced to capture the transition from conduction melting mode to keyhole melting mode in the above mentioned extended laser heat source model. The identification of these parameters is critical for accurately predicting the melt pool, especially for a complex L-PBF scanning process for intricate geometry component during which the parameters must be adaptively changed based on the local processing conditions. However, quantifying the explicit relationship between the local processing conditions and the melting mode transition parameters in the extended heat source models can be a difficult task. One viable approach to address this challenge is using data-driven surrogate models. Tapia et al. [24] and Xie et al. [25] has built Gaussian Process-based (GP) surrogate model based on single track experiments and thermal fluid flow simulation generated virtual data, respectively, to predict the melt pool depth given the laser process parameters. Ren et al.[26] used convolutional neural network (CNN)-based machine learning (ML) model to classify the intrinsic and perturbative keyhole oscillations from the operando high-speed x-ray imaging of the melt pool and detect the keyhole pore–generation. While those works are not used with a FE framework but as a direct numerical screening tool to identify the processing windows to avoid the keyhole mode, the data-driven surrogate model approach can be applied in this scope for identifying the parameters in the laser heat source model with melting mode transition feature.

In this paper, we propose a simple but effective method to extend the double ellipsoidal shape (Goldak) laser heat source model for the L-PBF process, in an effort to improve the accuracy of FE-based L-PBF process simulations. The proposed model introduces a laser penetration term, which is a function of laser parameters (power, scan speed, laser spot size), to the original Goldak model. Under high laser energy density, the model captures the high depth-to-width ratio keyhole mode melt pool profile; by decreasing the laser energy density, the model degenerate to the original Goldak formulation. The model parameters are determined using a Gaussian Process (GP)-based

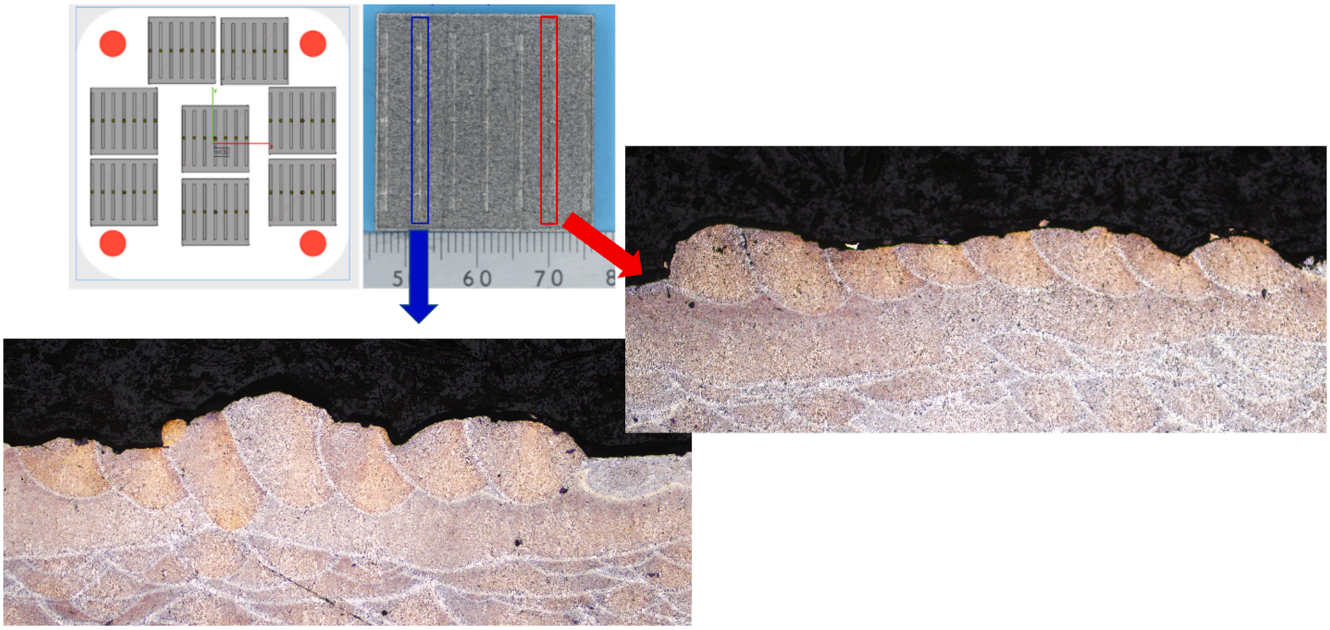

Fig. 2. Demonstration of ‘2D pad’ experiments and results.

Table 1
Measured ‘2D pad’ test melt pool width, depth and their standard deviation (SDV).

test ID	power (W)	speed (mm/s)	laser spot size (micron)	width (micron)	width-SDV (micron)	depth (micron)	depth-SDV (micron)
B1P2	250	1000	49.5	185.0	28.00	88.0	9.22
B2P7	250	1000	41	183.85	16.45	111.1	9.70
B2P5	250	1300	49.5	182	7.70	83	10.26
B5P7	250	1300	41	187.66	17.54	96	16.00
B4P3	250	1300	32.5	196.8	16.02	129.2	16.91
B6P3	300	1300	49.5	192.6	23.00	101.3	12.4
B2P2	300	1300	41	200.8	8.50	122.6	14.95
B4P4	300	1600	49.5	189.2	21.84	91.5	20.68
B6P4	300	1600	41	193.83	18.54	123.58	24.34
B6P1	300	1600	32.5	189.66	16.80	133.25	24.67
B2P3	325	1600	49.5	193.33	30.47	101.5	10.54
B1P6	325	1600	41	209.4	13.55	140.1	19.55
B5P6	325	1600	32.5	202.5	23.32	152.5	12.33
B5P3	375	1900	49.5	182	21.10	112.16	17.56
B4P6	375	1900	41	194.83	12.71	143	18.55
B3P6	375	1900	32.5	190.16	20.35	175.41	11.76

optimization method based on an open source Python MCMC sampler emcee code [27] under different L-PBF conditions. A series of AlSi10Mg ‘2D pad’ experiments are conducted under a broad range of processing parameters to demonstrate the process and validate the proposed model, with results demonstrating improved temperature predictions in handling complex L-PBF process conditions.

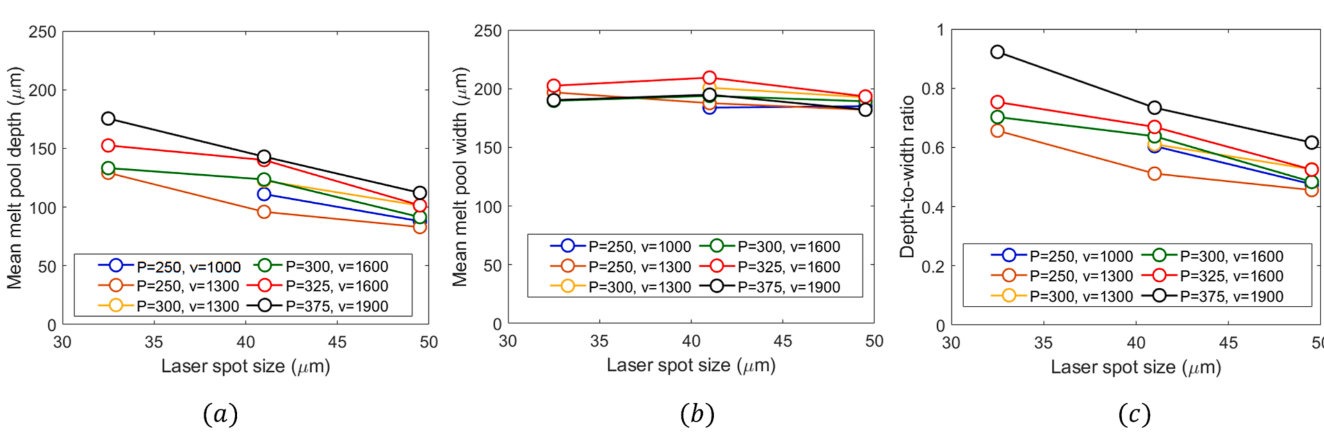

Fig. 3. 2D pad experiment results of melt pool (a) depth (b) width and (c) depth-to-width ratio.

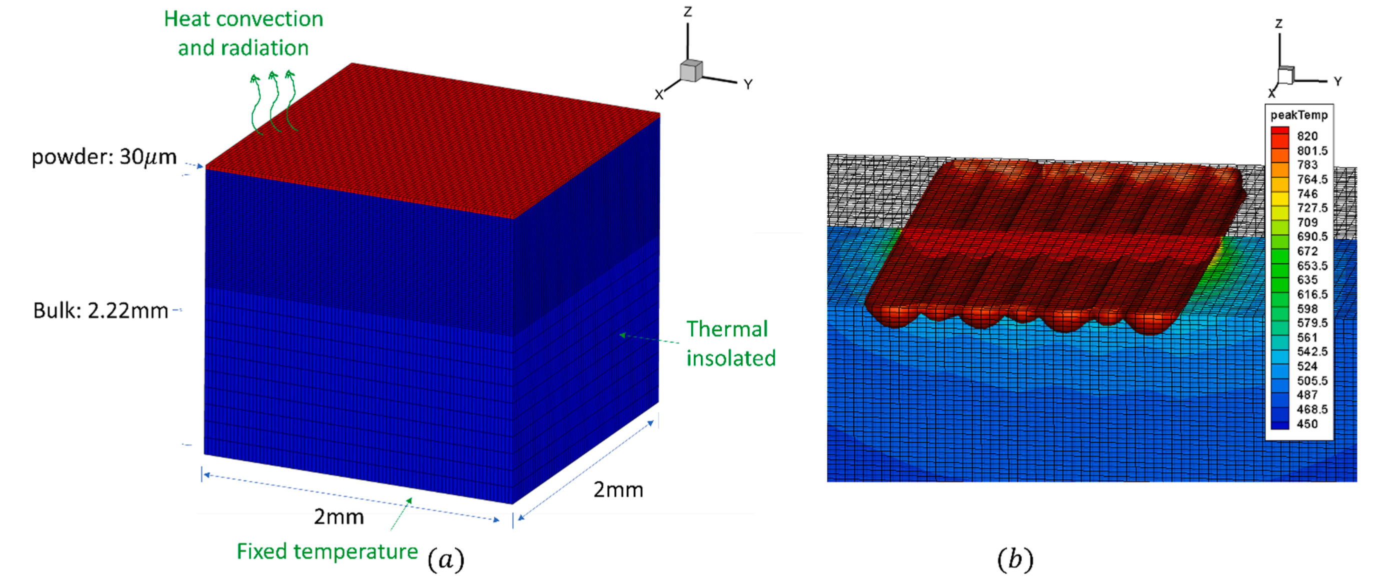

Fig. 4. (a) 2D pad FE simulation model setup and (b) illustration of resulting melt pool shape.

2. AlSi10Mg '2D Pad' experiment

We started by introducing the "2D Pad" experiments which are used to motivate and validate the model. The experiments were conducted by depositing AlSi10Mg tracks on "2D Pads" using SLM125 3D printer at Ford. On each pad, 7 tests were performed, and each test consisted of 7 or 8 tracks of bi-directional laser scans on one layer of AlSi10Mg powders located on the base. In each test, the laser controlling parameters for the each track were maintained as constants. The base, which has a "fish scale-like" structure (see Fig. 2), was printed prior to the tests with AlSi10Mg powders with the machine-default printing condition and laser printing direction rotates 67° every layer to minimize the potential residual stress. More than 50 tests have been conducted under various processing conditions, covering the laser power range from 250-375 W with the increment of 25 W, laser speed of 1000–2200 mm/s with the increment of 300 mm/s, and 3 laser focus offsets (0.67 mm, 2.67 mm and 3.67 mm). The powder layer thickness is 30 μm. The range of parameters were selected to cover both conduction melting mode and keyhole melting mode. The 2D pads were then cut, mounted, polished, and etched, to reveal the melt pool shapes, see example illustrated in Fig. 2.

The melt pool width and depth in the 2D pad tests corresponding to each set of printing parameters was measured using ImageJ Pro software. For each test, the mean melt pool depth, width and the standard deviation were characterized from all 7–8 tracks, and the results are listed in Table 1. The mean melt pool width and depth are plotted as a function of laser spot size in Fig. 3a-3c, respectively. The result revealed a clear trend that the melt pool depth increases with decreasing laser spot size, while melt pool width does not seem to be influenced by different laser spot size. Meanwhile, higher laser power leads to lager melt pool width and depth; and higher laser scan speed results in smaller melt pool width and depth, as expected [28]. These observations are consistent with existing literature reports [29]. To capture the distinctive geometric features of the conduction mode and keyhole mode, the depth-to-width (D/W) ratio of the melt pool is typically used in the literature [29,30] to distinguish the two melting modes. The critical D/W ratio is commonly accepted as 0.5, above which indicating keyhole mode. Fig. 3c shows the measured D/W ratio for each test. For a small laser spot size of 32.5μm, all measured melt pool D/W ratios are greater than 0.65 which correspond to keyhole melting mode. Larger laser spot size, smaller laser power and slower scan speed decrease the melt pool D/W ratio and shift the melting mode to conduction mode, as shown in Fig. 3c.

3. FE simulation based on double ellipsoidal (Goldak) heat source model

The FE simulations for the 2D pad experiments were conducted using ABAQUS/standard transient heat transfer analysis module [31] with user-defined heat fluxes subroutine DFLUX. The laser heat input was based on the Gaussian semi-ellipsoid heat source model from Goldak [32], which can be expressed as:

$$Q(x,y,z) = \begin{cases} \dfrac{6\sqrt{3}f_f\eta P}{a_f bc\pi\sqrt{\pi}}\exp\left(-3\left(\dfrac{d_x^2}{a_f^2}+\dfrac{d_y^2}{b^2}+\dfrac{d_z^2}{c^2}\right)\right) & for \quad d_x \geq 0 \\ \dfrac{6\sqrt{3}f_r\eta P}{a_r bc\pi\sqrt{\pi}}\exp\left(-3\left(\dfrac{d_x^2}{a_r^2}+\dfrac{d_y^2}{b^2}+\dfrac{d_z^2}{c^2}\right)\right) & for \quad d_x \leq 0 \end{cases} \tag{1}$$

here, $Q(x,y,z)$ is the volumetric heat flux at an integration point located at coordinates (x,y,z). The laser is assumed to move along x-axis. d_x, d_y, d_z are the components of spatial distance from this integration point to the moving laser spot center located on the top surface of the model, respectively. η is the laser energy absorptivity of the material, and P is the input laser power. a_f and a_r defines the semi-axis of the semi-ellipsoid for the front ellipsoid and rear ellipsoid parts, respectively; b defines the semi-axis transverse to the laser moving direction and c defines the depth of the semi-ellipsoid, respectively. For simplicity, $a_f = a_r = b = r$ is used in this work, with r being the laser spot size (radius). The coefficients f_f and f_r are introduced to define the fraction of heat deposited in the front and rear of the heat source, where $f_f + f_r = 2$. In addition, to ensure the continuity of equation (1) at$d_x = 0$, the constraints $\frac{f_f}{a_f} = \frac{f_r}{a_r}$ is required, and thus the values for f_f and f_r are determined.

The FE model for simulating the 2D pad experiments are plotted in Fig. 4a. As the direct simulation of the entire pad will lead to extremely large number of elements and intractable computation time, the FE model setup is for a representative volume element (RVE) cell of $2mm \times 2mm \times 2.25mm$ region. The chosen RVE size is consistent with the literatures [17,33,34] that study a single layer melt pool consisting of one or a few tracks. The RVE contains a 30 μm thick powder layer and a 2.22 mm thick base. The model is meshed into 502,897 eight-node linear hexahedron heat transfer elements, with finer $30\mu m \times 30\mu m \times 15\mu m$

Table 2
Thermal properties of AlSi10Mg [33].

Temperature (K)	298	473	673	820	870	1000
Density (bulk) (kg/m^3)	2650	2550	2400	2200	2000	1900
Thermal conductivity (bulk) ($W/(mK)$)	147	159	159	159	100	105
Specific heat capacity (bulk) ($J/(kgK)$)	739	797	838	922	1100	1000
Density (powder) (kg/m^3)	920	930	950	1000	–	–
Thermal conductivity (powder) ($W/(mK)$)	1.5	1.6	1.7	1.8	–	–
Specific heat capacity (powder) ($J/(kgK)$)	443	478	503	553	–	–

size elements at the top region and coarser $30\mu m \times 30\mu m \times 150\mu m$ elements at the lower region. Abaqus user-defined thermal material law subroutine UMATHT is used to define the thermal properties of powder phase and bulk phase. Abaqus user-defined heat transfer analysis subroutine FILM is used to apply the heat convection and radiation boundary conditions. Heat convection and radiation boundary condition are applied on the top surface. Considering the bottom and other surfaces are the boundaries between the modeled volume and the powder bed substrate, they are not exposed for heat convention or radiation. Following similar models in literature [33], fixed temperature boundary condition is applied on the bottom surface, while the rest surfaces are assumed to be thermally insolated. The elements for the base and the powder layer are assigned two different sets of material properties (bulk AlSi10Mg and powder AlSi10Mg), as listed in Table 2. The laser-powders interaction and the newly deposited layer is represented using a phase and properties switching method [19]. In the beginning of the simulation, the top layer elements of the mesh are 'flagged' (in the user defined subroutine UMATHT) as the powders and are assigned the temperature-dependent powder properties as listed in Table 2. When the temperature of any element on top layer increased above the solidus temperature, that element is flagged as a "bulk" element for the rest of the simulation and is assigned to "bulk" phase temperature-dependent thermal properties as listed in Table 2. It must be noted that this method has a drawback that it does not create side surfaces for the newly deposited tracks and therefore ignores the convection and radiation boundary conditions on those side surfaces. More sophisticated FEA method such as adaptively element activation method with adaptive boundary condition updates [35] can avoid such problem, which is not considered in this paper due to the implementation difficulty.

Fig. 4b illustrates the simulation predicted melt pool geometries by plotting peak-history temperature during the scanning of the seven tracks and the solidus temperature *iso*-surface, which corresponds to the surface of the melt pool boundaries.

For each 2D pad test in Table 1, multiple simulations were conducted based on varying calibratable parameters. It has been reported the vapor formation in keyhole melting mode would change the absorptivity as vapor has a different rate of absorption than the molten metal [36], and therefore both absorptivity η and laser heat source depth parameter c are considered as calibratable parameters.

However, the simulations based on original Goldak heat source model were unable to capture the 2D pad melt pool geometries for the results in keyhole melting mode. To further illustrate the discrepancy, Fig. 5 shows the simulation results for laser power $P = 375W$, scan speed $v = 1900mm/s$ and different laser focus offset (spot size). By setting absorptivity $\eta = 0.11$ and heat source depth parameter $c = 30\mu m$, the simulation is able to match the melt pool width, as shown in Fig. 5a. However, the melt pool depth was significantly underpredicted especially at smaller laser spot size, as shown in Fig. 5b. Increasing heat source depth parameter c did increase the predicted melt pool depth, however, this increase is quite marginal as shown in Fig. 5b where an

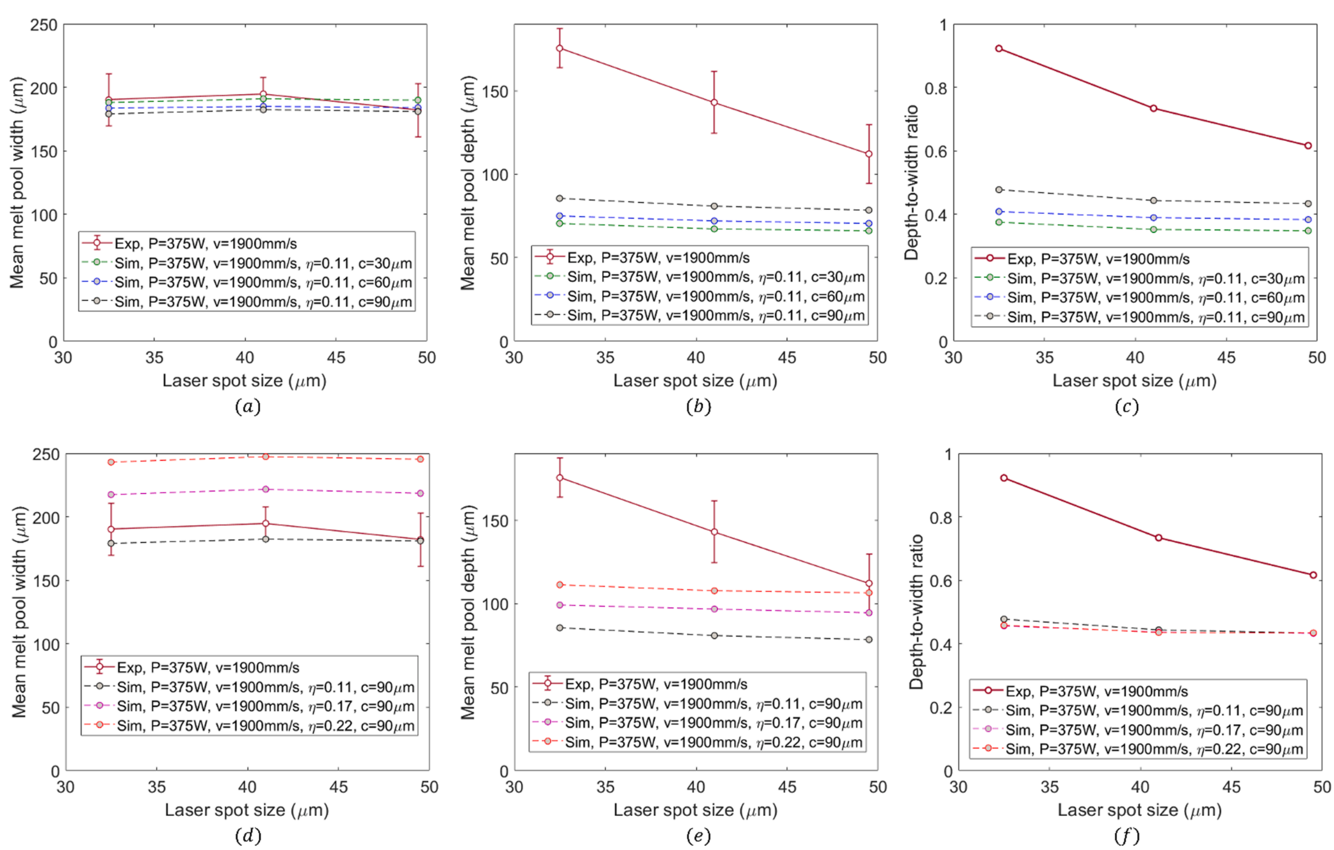

Fig. 5. Comparison of melt pool (a,d) depth, (b,e) width and (c,f) depth-to-width ratio between experiment measurements and simulation results based on Goldak laser heat source model with varying values of calibratable parameters (a-c) heat source depth c and (d-f) absorptivity η.

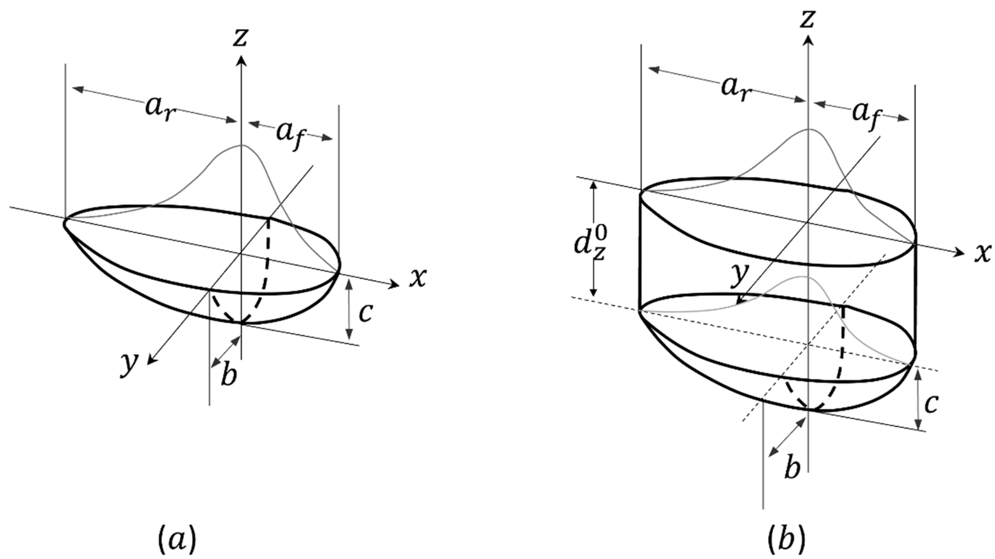

Fig. 6. Schematic of (a) double ellipsoidal (Goldak) heat source model, and (b) the extended Goldak heat source model.

increase of c from 30 μm to 90 μm only led to 12 μm $-$15 μm increase of predicted melt pool depth. Increasing absorptivity η, on the other hand, has a more obvious increase in melt pool depth (see Fig. 5e), but it also caused significant increase and overprediction of melt pool width (see Fig. 5d), leading to a nearly unchanged D/W ratio as shown in Fig. 5f. With all the attempts in calibrating absorptivity η and heat source depth parameter c, the predicted melt pool D/W ratio was always < 0.5, which corresponds to the conduction melting mode, as shown in Fig. 5c and 5f. These results indicate that calibrating η and c alone was unable to capture the keyhole melt pool formation in the simulation.

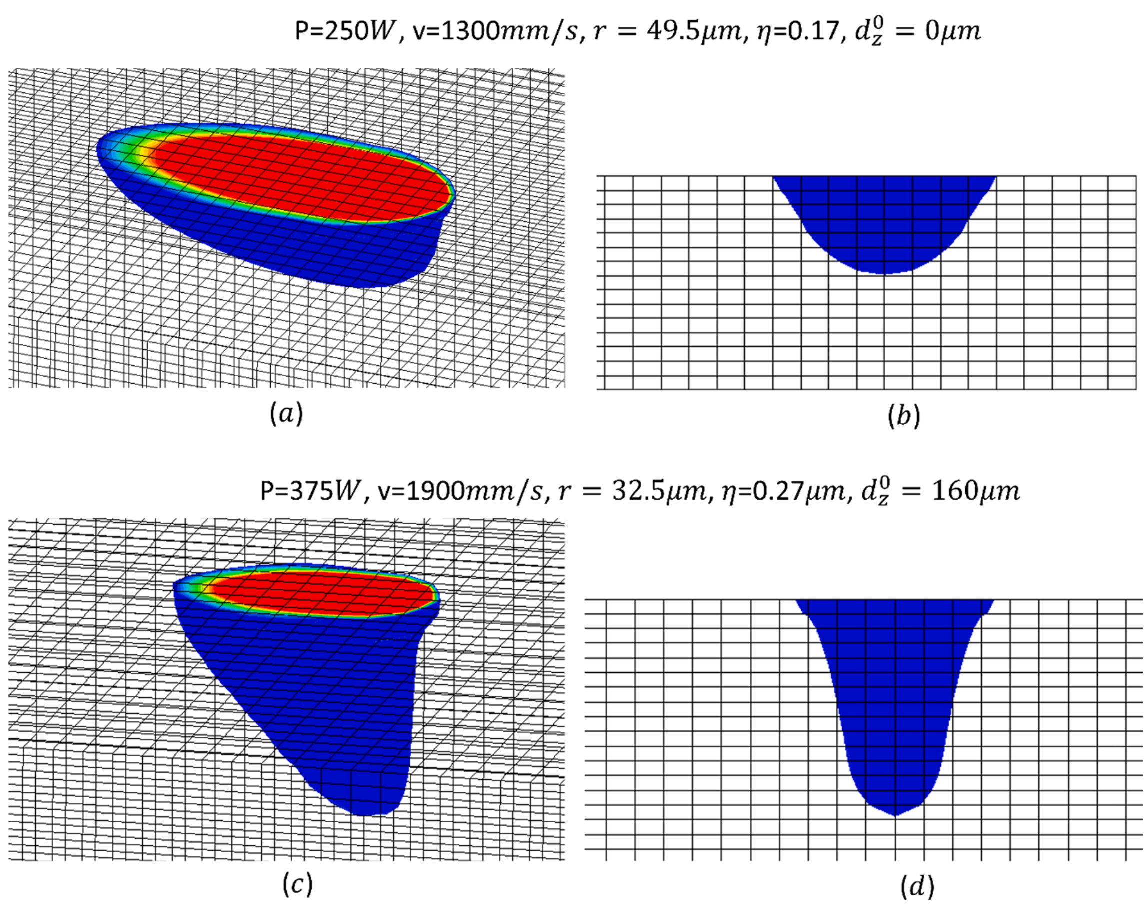

Fig. 7. Simulation results based on the proposed extended Goldak heat source model, which captures (a) conduction model melt pool shape and (c) keyhole mode melt pool shape. The melt pools are drawn using Tecplot software [37] with the *iso*-surfaces feature. The region with temperature below the solidus temperature (820 K) were removed in the visualization, and the regions with temperature above 1000 K were displayed in red color. (b) and (d) are the views along laser moving direction for (a) and (c), respectively.

4. Extended Goldak heat source model

In order to simulate the melt pool formation under a broad range of process parameters, we proposed a simple extension to the Goldak laser heat source model by introducing a physical parameter called laser penetration depth, d_z^0. The modified volumetric heat flux from the laser is expressed as:

$$Q(x,y,z)=\begin{cases}\dfrac{3f_{f/r}f_t\eta P}{a_{f/r}bd_z^0\pi}\exp\left(-3\left(\dfrac{d_x^2}{a_{f/r}{}^2}+\dfrac{d_y^2}{b^2}\right)\right) & for\ d_z<d_z^0\\ \dfrac{6\sqrt{3}f_{f/r}f_b\eta P}{a_{f/r}bc\pi\sqrt{\pi}}\exp\left(-3\left(\dfrac{d_x^2}{a_{f/r}{}^2}+\dfrac{d_y^2}{b^2}+\dfrac{\left(|d_z|-d_z^0\right)^2}{c^2}\right)\right) & for\ d_z\geq d_z^0\end{cases} \tag{2}$$

In equation (2), the parameters describing the shape and heat flux partition between the front and rear ellipsoid are independent of d_z and only functions of d_x, which is expressed as:

$$\begin{cases}f_{f/r}=f_f, & a_{f/r}=a_f \quad for \quad d_x\geq 0\\ f_{f/r}=f_r, & a_{f/r}=a_r \quad for \quad d_x\leq 0\end{cases} \tag{3}$$

The schematic of the extended heat source model is shown in Fig. 6b. d_z^0 can be physically interpolated as the laser penetration depth in keyhole mode when the vapor formation occurs with the recoil pressure pushing down the underneath melt flow. Numerically, d_z^0 controls the transition from conduction mode to keyhole mode, and is a function of laser power, scan velocity and laser spot size, i.e. $d_z^0 = d_z^0(P,\nu,r)$. When $d_z^0 = 0$, the equation (2) recovers the shape of conventional Goldak laser heat source model, and $d_z^0 > 0$ corresponds to the penetration of the laser beam into the melt pool due to the melt flow vaporization and recoil pressure. The determination of d_z^0 and absorptivity η as functions of laser power, scan velocity and laser spot size are described in section 4.1. f_t and f_b are the two additional parameters determining the partition of laser energy into the portion above and below d_z^0. The continuity of laser heat flux $Q(x,y,z)$ at d_z^0 requires $\frac{f_t}{d_z^0} = \frac{\sqrt{3}f_b}{\sqrt{\pi}c}$. In addition, the integration of equation (2) corresponds to the total absorbed power from laser, such that:

$$\int_{x=-\infty}^{x=+\infty}\int_{y=-\infty}^{y=+\infty}\int_{z=0}^{z=+\infty}Q(x,y,z)dxdydz=(f_t+f_b)\eta P=\eta P \tag{4}$$

which requires $f_t + f_b = 1$.

Fig. 7 shows the simulation predicted melt pool geometries in two case studies. The first case corresponds to a test with input laser power of 250 W, laser scan speed of 1300 mm/s, laser spot size of 49.5 μm. The calibratable parameters were set to $\eta = 0.17$ and $d_z^0 = 0\mu m$, and the simulation predicted a conduction mode melt pool geometry, as shown in Fig. 7a and 7b. The second simulation corresponds to a test with input laser power of 375 W, laser scan speed of 1900 mm/s, laser spot size of 32.5 μm. The calibratable parameters were set to $\eta = 0.27$ and $d_z^0 = 160\mu m$, and the simulation predicted a keyhole mode melt pool geometry, as shown in Fig. 7c and 7d. In both cases, the parameter c in the laser heat source model is fixed at $c = 30\mu m$. The simulation results demonstrate that even though the proposed modification in this work is rather simple, the extended Goldak heat source model achieves good versality in the sense that by choosing appropriate absorptivity η and penetration depth d_z^0 the model is able to capture both the conduction and keyhole melting modes.

4.1. Determination of d_z^0 and η using Gaussian process (GP) surrogate model

In order to accurately predict the melt geometries under varying test conditions with the extended Goldak heat source model, it is critical to establish the quantitative relationship between the calibratable parameters (absorptivity η and laser penetration depth d_z^0) and the laser process condition parameters (laser power P, scan speed ν and laser spot size r), i.e., to identify the functional form $\eta = \eta(P,\nu,r)$ and $d_z^0 = d_z^0(p,\nu,r)$. This functional relationship must be estimated from the available experimental data, but this task is not straight forward, the explicit forms of $d_z^0(P,\nu,r)$ and $\eta(P,\nu,r)$ are not yet known and need to be investigated with multi-physics particles-based or thermal fluid flow-based modeling and simulation methods. However, such physics-based methods are computationally too expensive to be explicitly used in an optimization-based calibration method.

Instead of pursuing a physics-based functional form for d_z^0 and η, we choose an alternative approach that adopts a data-driven statistical modeling method to construct a computationally efficient surrogate model for both desired variables. In this work, the surrogate model is constructed based on Gaussian processes (GP) modeling approach, with the details presented in this section. First, 50 FE simulations were conducted with distributed laser processing parameters and (d_z^0,η) model parameters in the design space, from which a GP surrogate model to emulate the FE-simulation results (accepts (P,ν,r,d_z^0,η) and predicts melt pool width) is created, as explained in subsection 4.1.1. Then, we construct another GP model to build the statistical relationship between processing parameters and (d_z^0,η), i.e., accepts (P,ν,r) and output $\left(d_z^0,\eta\right)$ Note that to differentiate the two surrogate models, we denote the first model as **GP FE-emulator** and second model as **GP physics parameter model**. The GP FE-emulator is used to efficiently identify the parameters in the likelihood function for GP physics parameter model, as described in section 4.1.2. The final output is the (d_z^0,η) for each set of laser process parameters (P,ν,r) using GP physics parameter model which is shown in Section 5.1.

4.1.1. Constructing GP FE-emulator

To simplify notation, we will hereafter refer to $d_z^0(P,\nu,r)$ simply as $f(\boldsymbol{x})$ and $\eta(P,\nu,r)$ as $g(\boldsymbol{x})$. Consider that N physical experiments have been performed and the width and depth of the melt pool have been measured at each condition, e.g. $\boldsymbol{w} = \{w_i\}_{i=1}^N$ and $\boldsymbol{h} = \{h_i\}_{i=1}^N$. For each experiment, there are controlled input variables $\boldsymbol{X} = \{\boldsymbol{x}_i\}_{i=1}^N$ where $\boldsymbol{x} = (P,\nu,r)$. Rather than estimating the desired functions, consider instead a simpler problem of simply estimating f_i and g_i at each individual experimental setting for $i = 1,\dots,N$. The following statistical model could be utilized to describe the data generating process,

$$\begin{aligned} w_i &= s_w(\boldsymbol{X},f_i,g_i)+\epsilon_i\\ h_i &= s_h(\boldsymbol{X},f_i,g_i)+\upsilon_i \end{aligned} \tag{5}$$

where ϵ_i and υ_i are experimental Gaussian noise with zero mean and unknown variances, and s_w and s_h are the finite element model predicted melt pool widths and heights. Note that here we assume that the finite element model is capable of exactly describing the data, and we did not incorporate a bias function which is sometimes included in the Bayesian calibration literature [38]. Point estimates for f_i and g_i can be estimated by minimizing the L_2 loss between w_i&s_w and h_i&s_h. Alternatively, a Markov Chain Monte Carlo (MCMC) sampler could be utilized to estimate probability densities of f_i and g_i from the statistical model. However, both cases are difficult as the finite element model is nonlinear, requiring iteration, and each simulation run is computationally expensive making direct iteration intractable. Hence, an alternative approach is needed to estimate the desired quantities.

A commonly used approach is to instead utilize a computationally efficient surrogate model to replace s_w and s_h. Establishing these surrogate models requires up-front computational cost accrued from running a fixed number of simulations over a suitable design. The inputs to the finite element model are parametric which include the prescribed experimental settings ($\boldsymbol{X}$) and unknown parameters d_z^0 and η; together

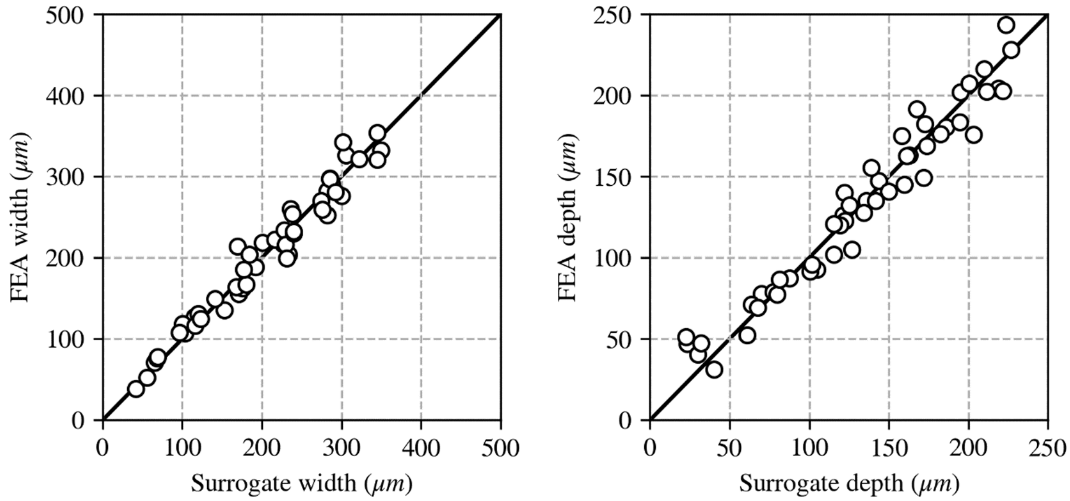

Fig. 8. Leave-one-out cross validation performance of the width and depth surrogate models.

$\boldsymbol{z} = \left(\boldsymbol{X}, d_z^0, \eta\right)$. In this work a M = 50 run maximum projection design was utilized for probing the finite element model [39]. This design criterion is desirable since it ensures good coverage, or spread, of settings over the input space and exhibits good properties for establishing surrogate models. The parametric finite element model can be emulated using a Gaussian Process surrogate model. Denote $\boldsymbol{y}$ as a single scalar emulated result (such as melt pool depth or width, the multiple outputs will be discussed later), over many runs the output of these computer experiments is $\boldsymbol{y} = \{y_i\}_{i=1}^{M}$, with M being the number of provided FE simulations. Under the assumption that these outputs are from a Gaussian Process the data can be jointly modeled according to the following statistical model,

$$\boldsymbol{y} \sim N\left(\boldsymbol{0}, \sigma_y^2 \boldsymbol{R}\right) \tag{6}$$

where the mean is assumed to be 0, σ_y^2 is the variance of the process, and data in $\boldsymbol{y}$ is correlated according to the correlation matrix $\boldsymbol{R}$. Each matrix entry captures the spatial correlation between two points $R_{ij} = \exp\left(-\frac{1}{2}\sum_{k=1}^{5}\theta_k^2 (z_{ik} - z_{jk})^2\right)$ where θ_k are roughness parameters that weigh the importance of different variables, e.g. the degree of spatial 'closeness' is different for P and V. Note that the statistical model contains no error term as the numerical simulation is deterministic: two repeated runs yield identical results (neglecting some floating point errors).

Note that this model contains hyperparameters $\boldsymbol{\theta}$ and σ_y^2 which must be estimated from the data. Point estimates of these values were obtained by maximizing the likelihood function for the statistical model (or equivalently minimizing the negative log-likelihood),

$$-\log\left(L\left(\boldsymbol{\theta}, \sigma_y^2\right)\right) = M\log(\sigma_y^2) + log(\det(R))$$
$$\sigma_y^2 = \frac{1}{m} y^T R^{-1} y \tag{7}$$

While there are ways to establish joint multiple-output Gaussian Process surrogates, we simply train two surrogates for each output (that is, in first surrogate model y is the melt pool width w, and in second surrogate y is depth h). For each surrogate model, predictions can be made using the following formula,

$$y(\boldsymbol{z}) = \boldsymbol{r}^T \boldsymbol{R}^{-1} \boldsymbol{y} \tag{8}$$

where $\mathbf{r}$ is a correlation vector $\mathrm{r_i}(\mathbf{z}) = \exp\left(-\frac{1}{2}\sum_{k=1}^{5}\theta_k^2 (z_{ik} - z_k)^2\right)$ and recall $\boldsymbol{z} = \left(\boldsymbol{X}, d_z^0, \eta\right)$ is an array containing the experimental settings $\boldsymbol{X}$ and d_z^0, η.

The results from above GP FE-emulator model are shown in Fig. 8. The leave-one-out cross validation (LOOCV) using a shortcut formula which exploits the structure in the correlation matrix is plotted to show the accuracy, which shows that surrogate models are indeed fairly accurate at emulating the finite element response.

4.1.2. *Determining the GP physics parameter model for (d_z^0, η) calibration*

Estimates of d_z^0 and η at each experimental setting ($\boldsymbol{x} = (P, v, \boldsymbol{r})$) can now easily be obtained by replacing the computationally expensive (actual FE-simulation based) functions s_h and s_w in Equation (5) and replacing them with the GP FE-emulator model estimated quantities. Assuming that measurement errors associated with width and depth observations are independent, the likelihood function can be described as,

$$p(w_i, h_i | \boldsymbol{x}_i, f_i, g_i) = \prod_{j=1}^{J} \frac{1}{\sigma_w\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma_w^2}(w_{ij} - \widehat{s}_w)^2\right) \prod_{j=1}^{J} \frac{1}{\sigma_h\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma_h^2}(h_{ij} - \widehat{s}_h)^2\right) \tag{9}$$

where $\widehat{s}_w = \widehat{s}_w(\boldsymbol{x}_i, f_i, g_i)$ and $\widehat{s}_h = \widehat{s}_h(\boldsymbol{x}_i, f_i, g_i)$ are the GP FE-emulator estimates of the finite element model width and height (from section 4.1.1), σ_w^2 and σ_h^2 are the variances for the experimental error in observing melt pool width and height, and with the introduction of some additional notation we allow for J experimental measurements of the quantities of interest, e.g. w_{ij} is the jth repeated width measurement in experiment i. In this work we use J = 5. Note, however, that the goal is to arrive at an estimate of the functions $f(\boldsymbol{x})$ and $g(\boldsymbol{x})$ and this expression only yields estimates of the discrete values f_i and g_i.

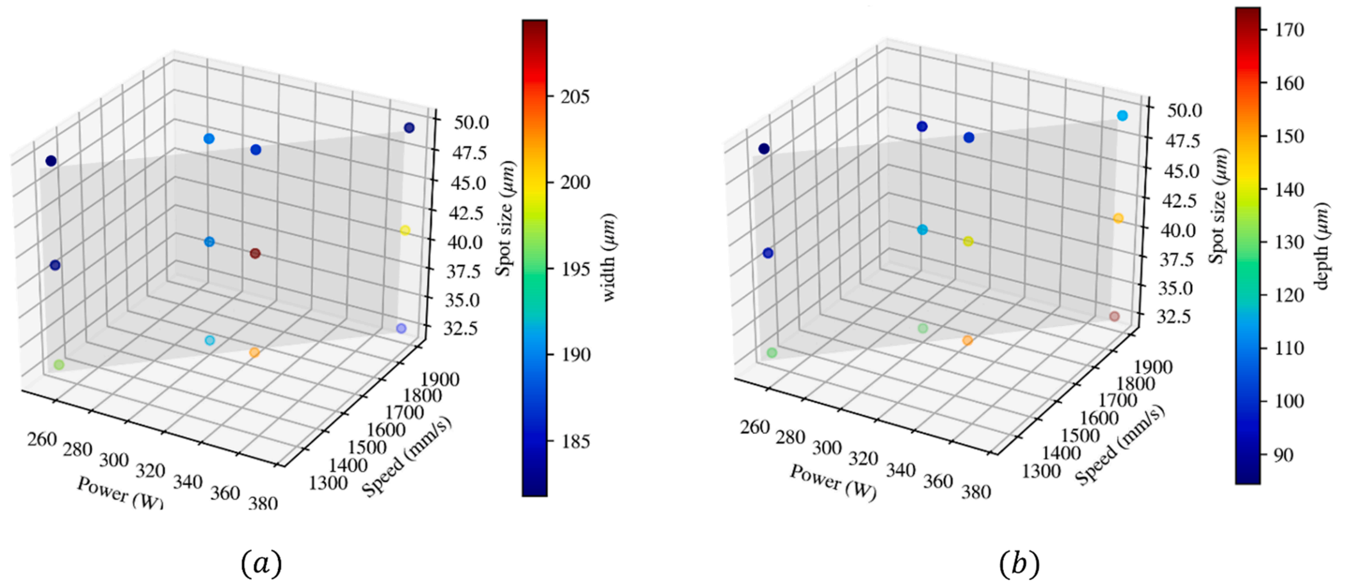

Fig. 9. Experimentally observed melt pool width and depth over 12 experimental settings. Approximate two-dimensional plane near which experiments we performed is shown.

The approach adopted is taken from prior work on modeling ion-channeling behavior in cardiac cells [38], where the authors addressed a similar problem: given many experimental settings, a functional form of a calibration parameter was desired. The key is to place a prior distribution on the desired functions which assumes that they come from a Gaussian Process,

$$
\begin{aligned}
p(f) &\sim N\left(\mu_f(\cdot), \sigma_f^2 r(\cdot,\cdot)\right) \\
p(g) &\sim N\left(\mu_g(\cdot), \sigma_g^2 r(\cdot,\cdot)\right)
\end{aligned}
\tag{10}
$$

where $\mu_f(\cdot)$ & $\mu_g(\cdot)$ are mean functions that capture global trends, σ_f^2 and σ_g^2 are the variance of the processes, and $r(\cdot,\cdot)$ a suitable spatial correlation function. In this work the mean function will be assumed to be

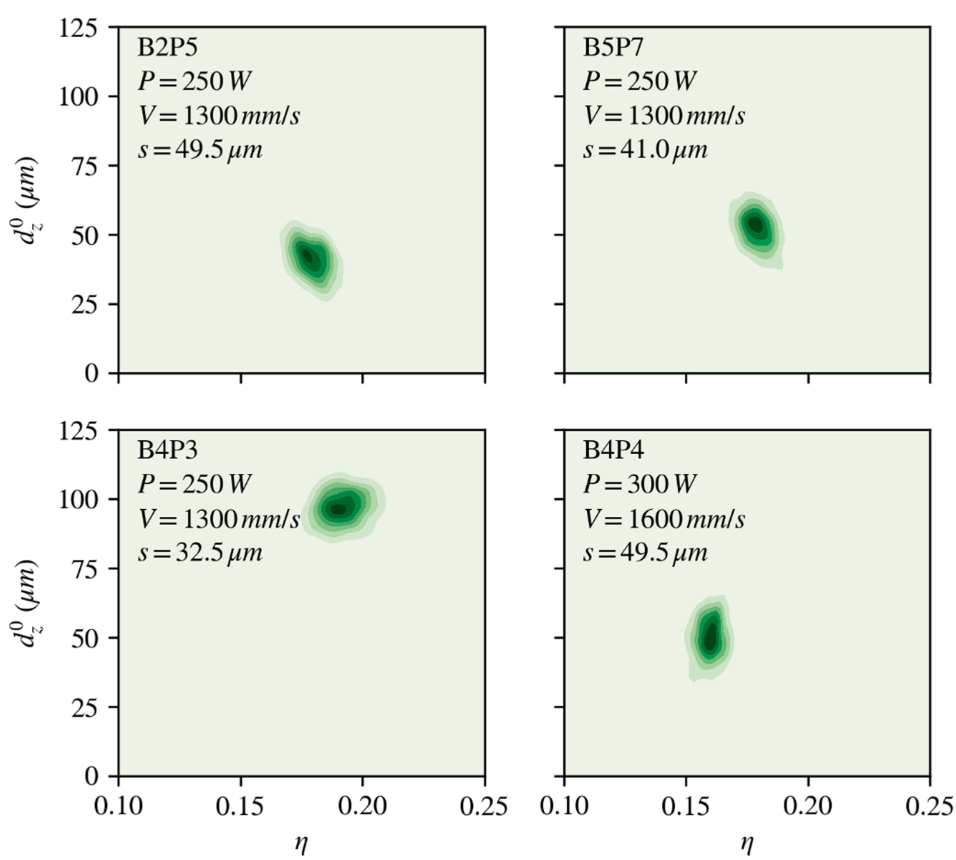

Fig. 10. Estimated densities for d_z^0 and η from the MCMC sampling for a few selected experiments.

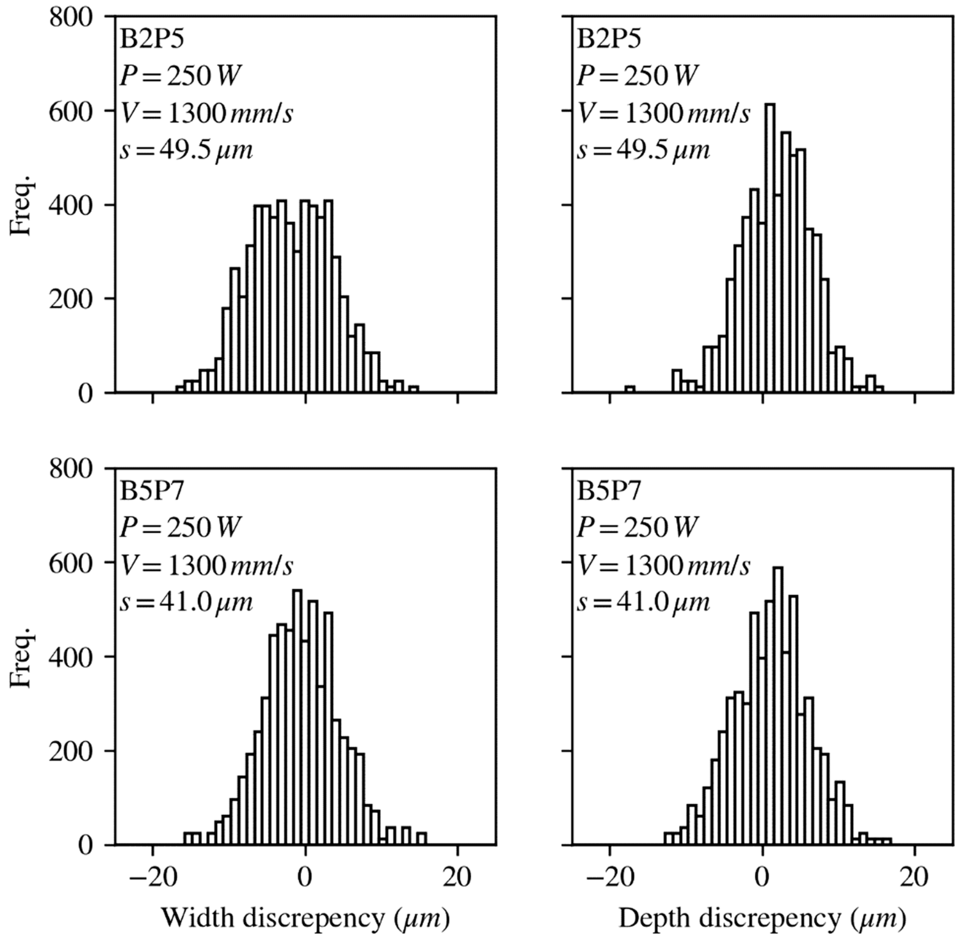

Fig. 11. Discrepancy between mean experimental width and depth and MCMC predicted width and depth for a few select experiments.

zero which is justified by mean-centering all the data prior to analysis. The correlation function is parameterized by roughness parameters ϕ in a similar way that was previously described. Consider all these hyperparameters to be lumped together into a quantity Φ. This prior is attractive since it is completely non-parametric; no assumption of the functional form is necessary. The Gaussian Process prior essentially allows for interpolation between discrete values of f_i and g_i. However, this flexibility requires that sufficient experimental and simulation data be available to confidently infer the unknown functions. The final statistical model, which is truly the GP physics parameter model that is desired, is

$$
\begin{aligned}
p(f, g|\boldsymbol{w}, \boldsymbol{h}, \boldsymbol{X}, \boldsymbol{f}, \boldsymbol{g}, \Phi) &\propto p(\boldsymbol{w}, \boldsymbol{h}|f, g, \boldsymbol{X}, \boldsymbol{f}, \boldsymbol{g}, \Phi)p(f|\boldsymbol{w}, \boldsymbol{h}, \boldsymbol{X}, \boldsymbol{f}, \boldsymbol{g}, \Phi)p(g|\boldsymbol{w}, \boldsymbol{h}, \boldsymbol{X}, \boldsymbol{f}, \boldsymbol{g}, \Phi) \\
p(f, g|\boldsymbol{w}, \boldsymbol{h}, \boldsymbol{X}, \boldsymbol{f}, \boldsymbol{g}, \Phi) &\propto p(\boldsymbol{w}, \boldsymbol{h}|\boldsymbol{X}, \boldsymbol{f}, \boldsymbol{g})p(f|\boldsymbol{f}, \Phi)p(g|\boldsymbol{g}, \Phi)
\end{aligned}
\tag{11}
$$

where in the second line all the unnecessary conditionally independent terms are eliminated e.g. $\boldsymbol{w}, \boldsymbol{h}$ are only dependent on the discrete values $\boldsymbol{f} = \{f_i\}_i^N$ and not the function f itself. Likewise the functions f and g can be constructed strictly from hyperparameters Φ and the discrete values $\boldsymbol{f}$ and do not dependent on experimentally measured width and height or the experimental process settings. The posterior distribution of the desired functions, f and g, is therefore constructed from the likelihood of the experimental data and priors placed on f and g. $p(f|\boldsymbol{f}, \Phi)$ and $p(g|\boldsymbol{g}, \Phi)$ can be easily computed from multivariate normal formulae. The unknown parameters associated with this model are: σ_w and σ_h, the experimental observation variances, N discrete values in f and g, and hyperparameters Φ. We obtained a discrete sample from these variables using Python MCMC sampler emcee [27]. No priors (uniform priors) were placed on f and g. Note that additional priors can be placed on any of the hyperparameters and we found it useful to place fairly tight priors on $\phi\ \mathcal{N}(1, 0.1^2)$. The choice of priors on roughness parameters is known to affect solution accuracy [40]. In earlier referenced work on cardiac cell modeling, the roughness parameter is fixed based on some physical knowledge of the system [38].

In addition, we employ one final data pre-processing step. In order to perform efficient experiments, instead of using all experimental data, only a subset of the (P, v, r) space was considered: an approximately two-dimensional plane with a near constant energy density, e.g. $\frac{P}{v} = const$, as can be seen in Fig. 9. The approach was chosen since we found that treating the problem in the original (P, v, r) space was problematic. We suspect that the difficulty has to do with the functional form of the correlation function $corr(\boldsymbol{x}_i, \boldsymbol{x}_j) = \exp\left(-\frac{1}{2}\sum_{k=1}^{3}\phi_k^2(x_{ik} - x_{jk})^2\right)$, where each component of x is statistically independently. However, in the experiments, P and v are usually varied in concordance. Hence, we

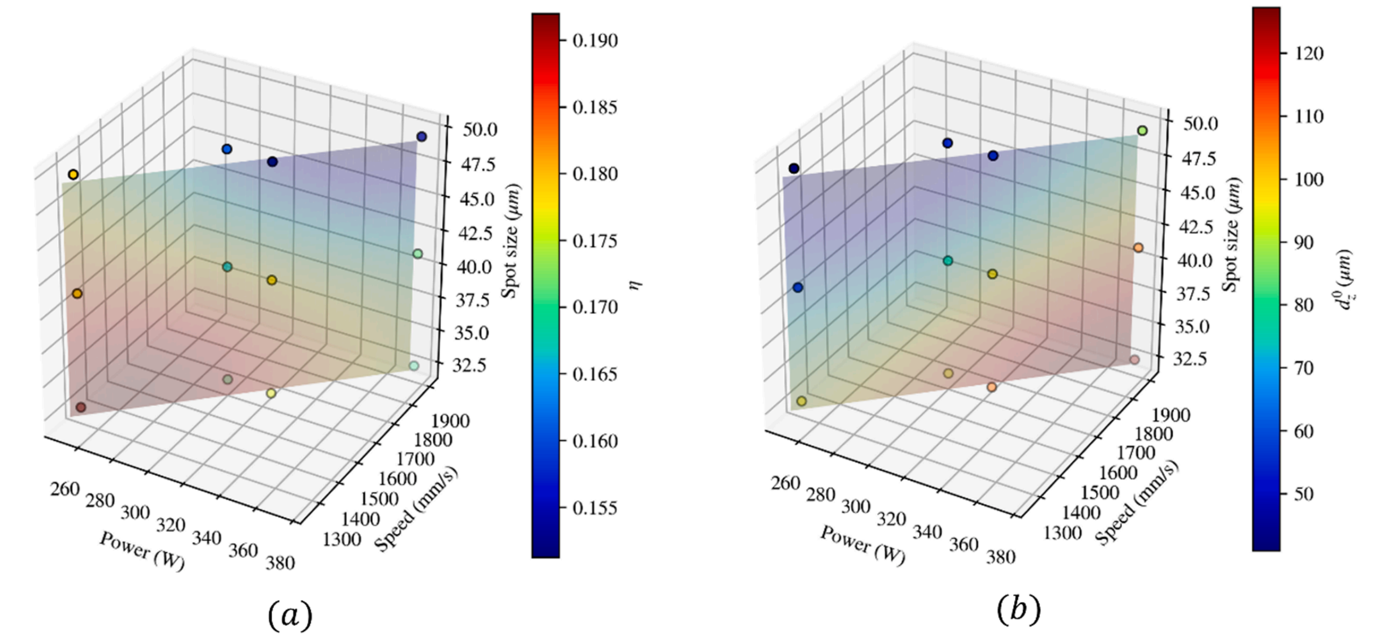

Fig. 12. Estimated mean functional values for (a) d_z^0 and (b) η over the experimental plane considered.

Table 3
Parameters η and d_z^0 for each 2D pad test determined using trained GP physics parameter model.

Power (W)	speed (mm/s)	laser spot size (μm)	Absorptivity η	penetration depth d_z^0 (μm)
250	1300	32.5	0.154	35
250	1300	41	0.185	45.2
250	1300	49.5	0.262	85
300	1600	32.5	0.165	39
300	1600	41	0.225	77
300	1600	49.5	0.235	90
325	1600	32.5	0.16	55
325	1600	41	0.22	101
325	1600	49.5	0.235	113
375	1900	32.5	0.172	83
375	1900	41	0.215	105
375	1900	49.5	0.235	135

performed principal component analysis (PCA) to obtain a transformation from (P, v, r)-space to a new coordinate system prior to performing the analysis. This can be done by a simple matrix multiplication $\boldsymbol{x}' = \boldsymbol{W}\boldsymbol{x}$ where $\boldsymbol{W}$ is obtained from PCA theory. This will also be useful for predicting functions f and g since predictions should be restricted to the plane defined by $\frac{P}{v} = const$, which is captured by the new PCA mapping $\boldsymbol{W}$.

Finally, with MCMC sampling, the parameters in the GP physics parameter model (equation (11) can be determined and the final η and d_z^0 values for each 2D pad experiment is presented in section 5.1.

5. Results

5.1. Using GP physics parameter model to identify η and d_z^0 for each 2D pad experiment

Using the trained GP physics parameter model for η and d_z^0, the probability densities for η and d_z^0 corresponding to different experimental settings (P,v,r) were identified and illustrated for four marginally varying experiments as shown in Fig. 10. Three of these experiments share identical power and speed with the laser spot size being the only variable. With decreasing spot size, it can be seen that the absorptivity increases and the parameter d_z^0 increases. In other words, as the spot energy becomes more focused, there is a tendency for the heat source to be driven further beneath the surface, which increases energy transfer from the laser source.

Fig. 11 illustrates the disparity between the mean measured weld pool width and depth and the predictions from the surrogate model in 4.1.2. Dispersion in the histogram is from uncertainty in the estimated parameters η and d_z^0 as well as uncertainty in other hyperparameters needed in the statistical model from Equation (8). The overall fit is good: predictions are generally within +/-10 μm with both depth and width predictions being generally symmetric and centered around 0.

Finally, the predicted mean values obtained from MCMC sampling of the posterior distribution for η and d_z^0 are shown in Fig. 12. In addition, a prediction of the functional quantities along a plane along $\frac{P}{v} = const$ is shown. The trend for d_z^0 agrees with physical intuition: with increasing speed/power, or decreasing spot size, there is an increasing tendency for keyholing to occur. The absorptivity behavior follows a similar trend; with decreasing spot size and decreasing speed or power, the absorptivity increases. The values of η and d_z^0 for each of the 12 2D pad experiment tests are listed in Table 3.

5.2. 2D pad simulation results using the extended Goldak laser heat source model

Finally, the 2D pad simulations were conducted based on the extended Goldak laser heat source model and the parameters identified from GP physics parameter model. The simulation predicted melt pool depth, width and D/W ratio for all 12 2D pad tests are compared with the corresponding experiment measurements in Fig. 13a-13c, respectively. In general, satisfactory results are obtained for most of the cases, except a slight overprediction of the melt pool depth in the case of P = 375 W, $v = 1900 mm/s$ and $r = 49.5 \mu m$, and a slight overprediction of melt pool width in the case of P = 300 W, $v = 1600 mm/s$ and $r = 32.5 \mu m$. The cross-sections of the seven tracks melt pool are also drawn in Fig. 14 for two cases, for visualization of the simulation predicted melt pool geometries under conduction mode and keyhole mode. As discussed in section 3, the satisfactory matching between the simulation results and experiments was not achieved using the original Goldak heat source model, showing the robustness of this new proposed method.

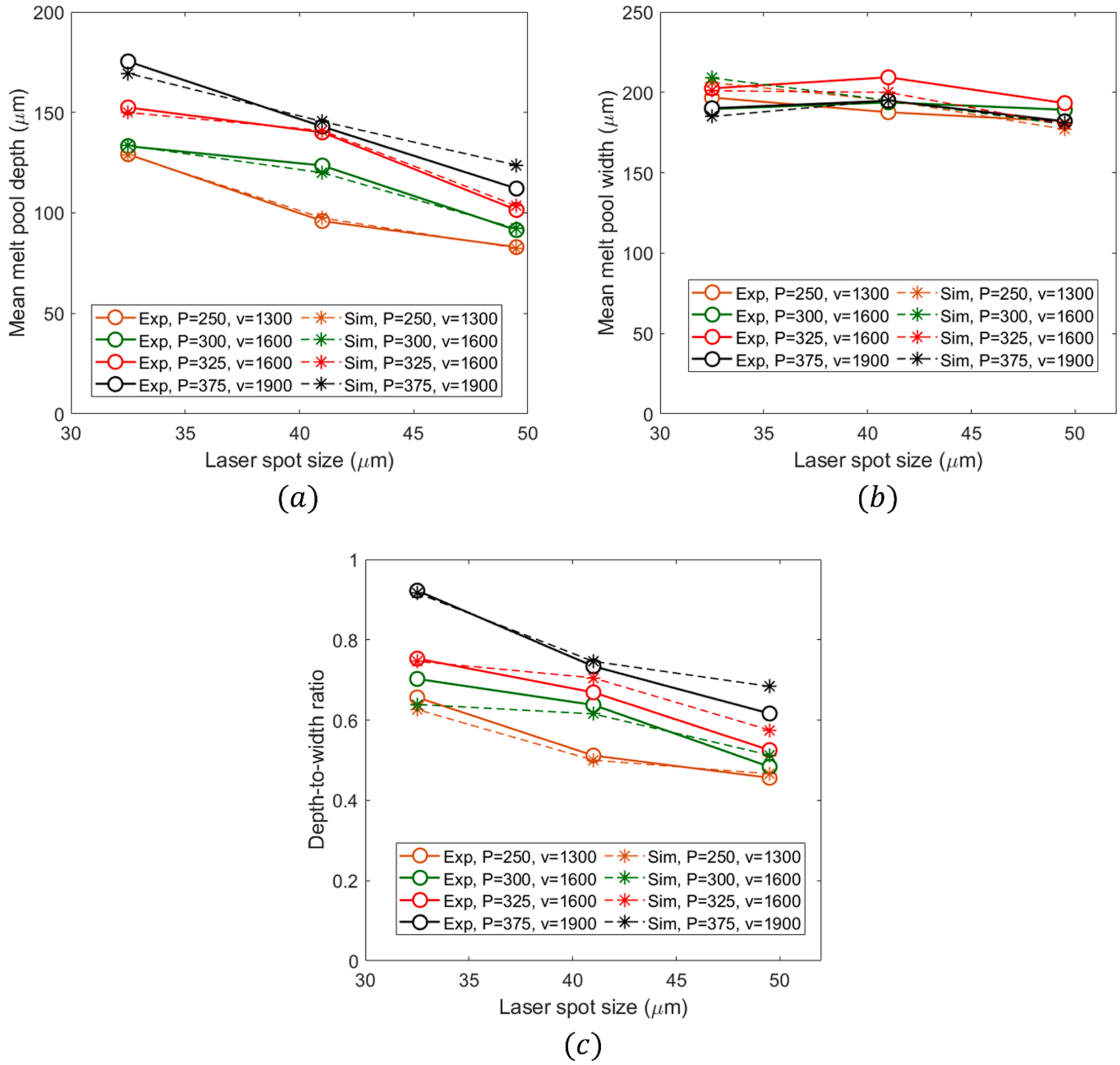

Fig. 13. Comparison of the simulation predicted and measured melt pool (a) depth (b) width, and (c) d/w ratio for each 2d pad tests.

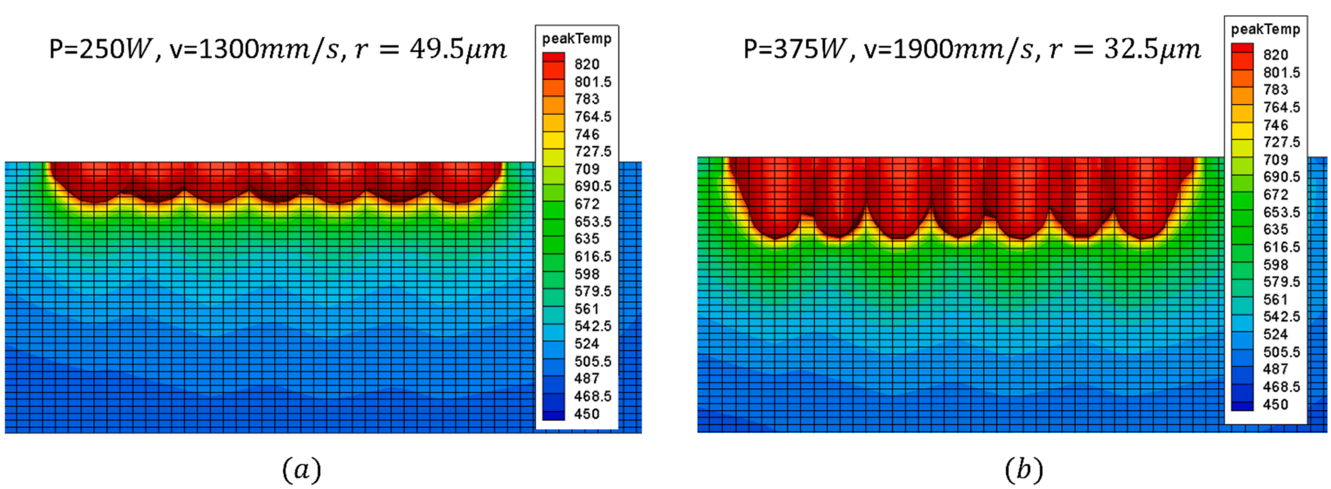

Fig. 14. Visualization of the simulation predicted melt pool geometries under (a) conduction mode and (b) keyhole mode.

6. Concluding remarks

In this paper, an extended double ellipsoidal (Goldak) laser heat source model is proposed for finite element thermal simulations to capture the transition between conduction melting mode and keyhole melting mode for different local laser processing conditions during laser powder bed fusion additive manufacturing. The model allows the laser heat source to penetrate into the melt pool, effectively capturing the physics of the high laser energy induced melt flow vaporization and recoil pressure, which lead to the keyhole shape melt pool formation. The calibratable model parameters, laser energy absorptivity η and penetration depth d_z^0, are treated as functions of laser process parameters including laser power, scan speed and laser focus offset (spot size). The identification of calibratable parameters according to various laser conditions are based on constructing two Gaussian process surrogate models (GP FE-emulator and GP physics parameter model), trained with virtual FE experiments. Various "2D pad" AlSi10Mg L-PBF experiments under a wide range of laser power, scan speed, and laser focus offset were conducted to validate the proposed model. The measured 2D pad test melt pool dimensions presents both conduction melt pool shapes and keyhole melt pool shapes, depending on the laser condition. Using the proposed model, the FE simulation predicted melt pool results showed good comparison with experiments for all cases, which was not achieved with original Goldak heat source model. The proposed model and method are thus expected to improve the reliability of FE simulations, leading to higher accuracy process design tool for avoiding keyhole melting-induced manufacturing defects in complex L-PBF process.

CRediT authorship contribution statement

Jiahao Cheng: Writing – original draft, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Yang Huo:** Writing – review & editing, Investigation, Formal analysis, Data curation, Conceptualization. **Patxi Fernandez-Zelaia:** Writing – review & editing, Validation, Software, Methodology, Investigation. **Xiaohua Hu:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Mei Li:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization. **Xin Sun:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

The raw/processed data required to reproduce these findings cannot be shared at this time due to technical or time limitations.

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